

[This question paper contains 8 printed pages.]

Your Roll No.....

Sr. No. of Question Paper : 5811

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Unique Paper Code : 2352012302

Name of the Paper : Riemann Integration

Name of the Course : B.Sc. (H) Mathematics

Semester : III (DSC)

Duration : 3 Hours

Maximum Marks : 90

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Attempt **all** questions, selecting any **three** parts from each question.
3. All questions carry equal marks.
4. Use of calculator is NOT allowed.

P.T.O.

1. (a) Calculate the upper and lower Darboux integrals for the function $f(x) = x^3$ on the interval $[0, b]$ and show that its integral value is $\frac{b^4}{4}$.

- (b) Let f be a bounded function on $[a, b]$. If P is a partition of $[a, b]$ then

$$m(b-a) \leq L(f, P) \leq U(f, P) \leq M(b-a)$$

where $M = \sup\{f(x) : x \in [a, b]\}$ and $m = \inf\{f(x) : x \in [a, b]\}$. Verify the inequality for the function $f(x) = 3x + 1$ defined on $[0, 3]$ and the partition =

$$P = \left\{0, \frac{1}{2}, 1, 2, 3\right\} \text{ of } [0, 3].$$

- (c) Let f be a bounded function on $[a, b]$. Show that f is integrable over $[a, b]$ if and only if for each $\epsilon > 0$, there exists a partition P of $[a, b]$ such that $U(f, P) - L(f, P) < \epsilon$.

- (d) Define Riemann sum and Riemann integrability of a bounded function f on $[a, b]$. Let f be a bounded integrable function on $[a, b]$ and $\{S_n\}$ be a sequence of Riemann sums with corresponding partitions P_n satisfying $\lim_{n \rightarrow \infty} \text{mesh}(P_n) = 0$. Show that the

sequence $\{S_n\}$ converges to $\int_a^b f$.

2. (a) Show that if a function f is integrable on interval $[a, b]$, then f is integrable on every interval $[c, d] \subseteq [a, b]$.

- (b) Let f be a bounded function on $[a, b]$. If Q is a refinement of a partition P of $[a, b]$, then show that $U(f, Q) - L(f, Q) \leq U(f, P) - L(f, P)$.

- (c) Prove that a bounded function f on $[a, b]$ is Riemann integrable if it is Darboux integrable. Also compare their corresponding values of the integrals.

- (d) Let f and F be functions on $[0, 1]$, given by $f(x) = \cos(x)$ for all $x \in [0, 1]$ and

$$F(x) = \begin{cases} 0, & \text{for } 0 \leq x < \frac{\pi}{6} \\ 1, & \text{for } \frac{\pi}{6} \leq x \leq 1 \end{cases}$$

Show that f is F -integrable and $\int_0^1 f dF = \frac{\sqrt{3}}{2}$.

3. (a) Let f be an integrable function on $[a, b]$ and there exists $\lambda > 0$ such that $|f(x)| \geq \lambda \forall x \in [a, b]$. Show that $1/f$ is integrable over $[a, b]$.

- (b) Let f be a continuous function on $[a, b]$ such that

$f(x) \geq 0$ for all $x \in [a, b]$. Show that $\int_a^b f = 0$

if and only if $f(x) = 0$ for all $x \in [a, b]$.

- (c) Let f be a function defined on $[a, b]$ and $c \in (a, b)$ be any point. Show that if f is integrable on $[a, c]$ and $[c, b]$, then f is integrable on $[a, b]$ and

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

- (d) State and prove intermediate value theorem for integrals. Suppose f and g are continuous functions

on $[a, b]$ such that $\int_a^b f = \int_a^b g$. Prove there exists x in (a, b) such that $f(x) = g(x)$.

4. (a) Define piecewise continuous function. Show that every piecewise continuous function on $[a, b]$ is integrable.

- (b) State and prove Fundamental Theorem of Calculus I. Using the same, evaluate

$$\int_0^{\pi} x \cos x \, dx.$$

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- (c) Show that if f and g are integrable functions over $[a, b]$ and if $f(x) \leq g(x)$ for $x \in [a, b]$, then

$$\int_a^b f \leq \int_a^b g. \text{ Hence deduce that}$$

$$\int_{-2\pi}^{2\pi} |x^2 \sin^8(e^x)| dx \leq \frac{16\pi^3}{3}.$$

- (d) Show that if f is a continuous real-valued function on $[a, b]$ satisfying $\int_a^b f(x)g(x) dx = 0$ for every continuous function g on $[a, b]$, then $f(x) = 0$ for all x in $[a, b]$.

5. (a) Use disk and washer method to find the volume of the solid generated when the region enclosed by $y = \sqrt{x+1}$, $y = \sqrt{2x}$ and $y = 0$ is revolved about x -axis.

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- (b) Use cylindrical shell method to find the volume of the solid generated when the region enclosed by $y = x^3$, $y = 1$, $x = 0$ is revolved about the line $y = 1$.

- (c) Find the exact arc length of the curve $x = \frac{y^4}{8} + \frac{1}{4y^2}$ from $y = 1$ to $y = 4$.

- (d) Find the area of surface generated by revolving the curve

$$x = t, y = 2t^2, 0 \leq t \leq 1$$

about y -axis.

6. (a) Discuss the convergence or divergence of the following integrals.

$$(i) \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$$

$$(ii) \int_1^{\infty} \frac{1}{x^3} dx$$

P.T.O.

(b) Test for convergence

$$\int_0^{\infty} \frac{1}{\sqrt{x}e^x} dx$$

(c) Suppose $\int_a^b f$ is an improper integral of type I.

Show that if $\int_a^b |f|$ converges, then so does $\int_a^b f$.

(d) Show that $\int_0^1 t^{p-1} (1-t)^{q-1} dt$ converges for positive values of p and q .